

Paper OOL-4: Interfacial Water, Proton/Ion Coupling, and Proto-Chemiosmosis: Closing the Electron-Proton Loop

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Abstract

Electron transport is incomplete without charge compensation, proton/ion motion, and hydrated interfaces. This paper gives the proton side of the geochemical coupling layer: mineral-water boundaries, surface charge, Grotthuss-like transport, pH gradients, and pore geometry. CDFD treats interfacial water as a conditional coupling layer, not as an exotic permanent phase. The central test is whether electron and proton/ion currents overlap in space and time strongly enough to support work-like chemistry.

1 Introduction

Chemiosmosis stores energy in electrochemical potential differences, chiefly transmembrane proton gradients in modern cells. Before lipid bilayers, mineral–water interfaces could still sustain localized pH and potential differences [6, 7]. Paper OOL-4 therefore asks: **how should proton transport be represented next to electron transport in a single constraint-driven framework?**

Carried gap from Paper 3: distributed electrons require a matching proton subsystem for redox coupling and vectorial chemistry.

Status: Inferred

Capstone channel mapping. Paper 12 maps structured interfacial water to the proton/ion coupling channel σ_p . This paper supplies the lower-level mechanism: electron transport alone is incomplete unless charge compensation, proton motion, or ion gradients can be coupled to the redox field. The conductance proxies relate to the local resistances as $\sigma_e \sim 1/C_e$ and $\sigma_p \sim 1/C_p$.

2 Model and Assumptions

2.1 Shared notation

Electron and proton channels each admit flux laws

$$\mathbf{J}_e = -\frac{1}{C_e} \nabla \Phi_e, \quad \mathbf{J}_p = -\frac{1}{C_p} \nabla \Phi_p, \quad (1)$$

with nonnegative resistances C_e, C_p . The CDFD adaptive ratios are $\Psi_{s,e} = \Phi_e/C_e$, $\Psi_{s,p} = \Phi_p/C_p$. In tri-regime language (Paper 12), $\sigma_e \sim 1/C_e$ and $\sigma_p \sim 1/C_p$ are conductance proxies.

Status: Derived

2.2 Bulk and interfacial water

Liquid water is a fluctuating hydrogen-bond network. Near polar mineral surfaces, dipole orientation, density, and mobility differ from bulk [3]. Such interfacial order can enhance or gate proton hopping depending on surface charge and ionic strength.

Status: Inferred

2.3 Grotthuss mechanism

Proton displacement along hydrogen bonds proceeds via bond rearrangement (classical and quantum contributions) with effective mobility exceeding that of hydrodynamic diffusion [1]. Large-scale modeling often collapses this physics into an effective conductivity tensor; here we use a scalar disorder parameter.

Status: Inferred

2.4 Proto-chemiosmotic loop (hypothesis)

Coupling requires correlated electrochemical potentials,

$$\Delta \Phi_e \sim \Delta \Phi_p + \text{offset}, \quad (2)$$

so that redox exergonicity can pump protons or vice versa. Mineral–water–solute microenvironments may realize primitive analogs before lipid encapsulation [2].

Status: Hypothesis

2.5 Toy scalar feedback model (script)

`supplementary.o01_04.py` tracks disorder $C_p \geq C_{\min}$ with $C_{\min} = 0.05$, driven by imposed gradient G and throughput $J_p \approx G/C_p$. Feedback reduces C_p when J_p is large, modeling flow-aligned ordering. This is not molecular dynamics.

Status: Hypothesis

3 Main Results

3.1 Conceptual

Structured water provides the proton counterpart to mineral electron pathways. Constraint fields C_p encode network disorder; reductions in C_p map to higher effective Grotthuss throughput in the phenomenology.

Status: Inferred

4 Numerical Verification

Companion script: `supplementary_ool_04.py`.

- Final disorder $C_p = 0.050$ for imposed gradients $G \in \{0.5, 1.0, 3.0, 5.0\}$.
- Final proton-flow values: 38.969, 106.238, 360.965, 613.482.

Strong driving collapses the model to its disorder floor and yields steeply increasing throughput—consistent with “ordering under flux” but not proving a specific geological site.

Status: Derived

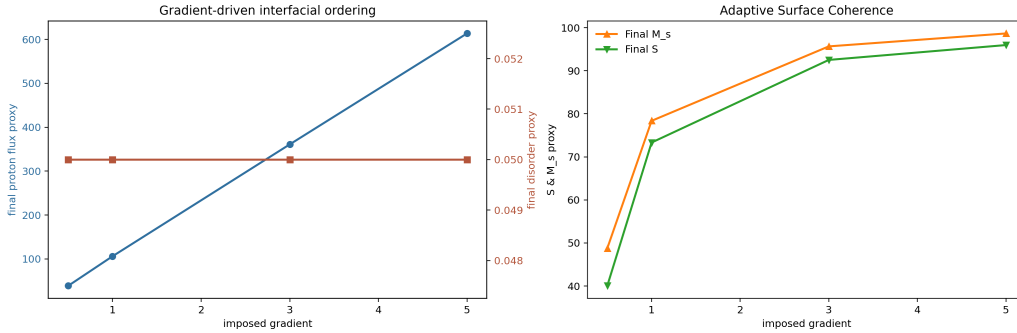


Figure 1: Structured-water gradient diagnostic. Stronger imposed gradients drive the toy disorder variable toward its lower bound and increase the proton-flux proxy. The calculation motivates an interfacial proton-coupling role, but it is not molecular dynamics.

5 What “Structured Water” Means Here

The phrase does not require a permanent exotic phase of water. In this paper it means transient orientational and hydrogen-bond organization near charged, mineral, or confined interfaces. Such organization can alter proton hopping, local dielectric response, ion mobility, and residence time. In the OOL chain, this function is needed because electron transport alone cannot perform chemical work unless charge compensation and proton chemistry remain nearby.

Status: Inferred

6 Coupling to Early Chemiosmosis

Modern cells use membranes, pumps, and protein complexes to manage proton gradients. The early analogue need not be a membrane protein. A mineral pore can separate alkaline and

acidic fluids; interfacial water can provide proton-relay structure; redox-active minerals can supply electron flow. The dependency is therefore:

$$\mathbf{J}_e \text{ available} + \mathbf{J}_H \text{ co-localized} \Rightarrow \text{redox chemistry can become work-like.} \quad (3)$$

Paper 8 then asks how long these coupled products stay localized before dilution destroys the loop.

Status: Inferred

7 Proton Gradients: Useful but Contested

The vent-gradient literature is central but not settled. The alkaline-vent view argues that natural pH differences and thin inorganic barriers could provide primitive chemiosmotic structure before enzymes and lipid membranes [6, 10, 5]. The critique is that natural gradients may dissipate, may not couple to early chemistry efficiently, or may be too different from biological membranes to do the required work [4]. CDFD does not hide this disagreement; it reframes the question as a measurable coupling problem:

$$\mathcal{C}_{ep} = \frac{\int_{\Omega} |\mathbf{J}_e \cdot \mathbf{J}_H| dV}{\left(\int_{\Omega} |\mathbf{J}_e|^2 dV\right)^{1/2} \left(\int_{\Omega} |\mathbf{J}_H|^2 dV\right)^{1/2}}. \quad (4)$$

The exact normalization can be improved, but the aim is clear: electron and proton/ion currents must overlap in space and time. A pH gradient alone is insufficient if it is geometrically decoupled from redox chemistry.

Status: Hypothesis

8 Surface Charge, Screening, and Pore Geometry

The phrase “structured water” becomes more defensible when tied to ordinary interfacial physics. Charged mineral walls organize counterions and water dipoles over a finite electrostatic length scale. In dilute electrolyte the Debye length can be written schematically as

$$\lambda_D = \sqrt{\frac{\epsilon k_B T}{2 N_A e^2 I}}, \quad (5)$$

where I is ionic strength. The exact expression depends on solution composition, but the implication is simple: surface effects dominate only when pore radii, wall separation, hydration films, and ionic strength place a meaningful fraction of the water volume inside the interfacial zone.

Thus the claim is not that all water in a vent or pool becomes a coherent proton medium. The stronger claim is geometric:

$$\frac{A_{\text{surface}} \lambda_D}{V_{\text{pore}}} \ll 1 \Rightarrow \text{interfacial transport can influence bulk chemistry.} \quad (6)$$

Small pores, charged walls, thin films, and mineral lamellae make this ratio larger. Open water

makes it smaller. This connects Paper 4 directly to Paper 8: compartments are not only retention devices; they also increase the fraction of chemistry exposed to interfacial proton/ion physics.

Status: Derived guardrail

9 Electron–Proton Coupling Ledger

The minimum requirement for proto-chemiosmotic relevance is co-localization of three layers:

Layer	Failure if absent	Measurable proxy
Electron transport	Redox drive cannot reach reaction interface	Source–sink conductance or redox-state change
Proton/ion pathway	Charge separation dissipates or stalls chemistry	pH persistence, proton mobility, ion-current overlap
Retaining geometry	Products and gradients wash out	Residence time, concentration enrichment, pore cycling

Paper 4 owns the middle row, but it only works in the full sequence. If electron transport is missing, proton gradients are uncoupled acidity. If retention is missing, proton coupling becomes a transient flash. If the proton path is missing, electron transfer becomes charge-imbalanced chemistry with poor work extraction.

Status: Derived synthesis

10 Water, UV, and Surface Boundary Conditions

Interfacial water also controls which photochemical routes remain plausible. UV transmission depends strongly on water depth, dissolved ions, suspended minerals, and atmospheric filtering [9, 8]. A shallow surface pool, a mineral pore, and a hydrothermal chimney do not expose chemistry to the same radiation or hydration regime. This is why structured water is not only a proton-wire concept; it is part of the environmental boundary condition that decides whether chemistry is hydrated, dehydrated, shielded, or irradiated.

Status: Inferred

11 Interpretation

The results support treating interfacial water as an active participant, not a passive solvent [6]. Quantitative predictions require atomistic or at least continuum Poisson–Nernst–Planck treatments absent here.

Status: Inferred

12 Limitations and Open Problems

- Scalar disorder cannot capture anisotropic hydrogen-bond chains.
- Floor saturation may dominate at high drive, masking richer physics.

- Coupling constants between Φ_e and Φ_p are not identified from data.

Status: Open

13 Conclusion

Paper OOL-4 closes the electron–proton loop at the field-description level and motivates interfacial structuring as a constraint problem. Paper OOL-8 examines how porous compartments and adaptive boundaries change persistence thresholds against dilution.

Status: Inferred

14 Reproducibility Note

`python supplementary_ool_04.py`

The script writes compact reproducibility outputs under `outputs/paper_04/`.

References

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